Global Fashion Retail Analytics

End-to-End Retail Data Analysis Using SQL and Power BI

Capstone Project

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Python / Pandas / MySQL / SQL / Power Query / Power BI and DAX

Project Objective

To analyze global fashion retail data using **MySQL and Power BI** and convert raw transactional data into meaningful business insights

- Evaluate sales, revenue, returns, and basket performance
- Identify high-performing products, categories, stores, and regions
- Understand customer segments, purchasing patterns, and retention
- Measure the effectiveness of discounts and promotions
- Build interactive dashboards with filters and drill-down analysis
- Support decisions related to product assortment, promotions, customer retention, and return reduction



Dataset Overview

- The dataset represents a global fashion retail business operating across multiple countries and currencies.
- It contains more than **6.4 million transaction records**.
- The available data covers customer purchases, product details, stores, discounts, sales transactions and returns.
- Both **Sale** and **Return** transaction types are included.
- Important fields include transaction date, invoice ID, customer ID, product ID, store ID, quantity, unit price, discount, line total, payment method and currency.
- The datasets were imported into **MySQL** and connected through common identification fields.



Project Workflow

- **Data Collection**
  Collected customer, product, store, transaction, discount, and return datasets.
- **Data Import**
  Used Python and Pandas to load CSV files into MySQL.
- **SQL Analysis**
  Used joins, CTEs, views, aggregations, and window functions to answer business questions.
- **Data Cleaning**
  Handled missing values, duplicates, incorrect formats, and inconsistent records.
- **Data Modelling**
  Created table relationships and a Calendar table for time-based analysis.
- **Power BI Dashboard**
  Connected MySQL through ODBC and created DAX measures, interactive visuals, filters, and drill-downs.
- **Insights and Recommendations**
  Evaluated sales, returns, geographic performance, categories, promotions, and customer behaviour.

- **Tools Used:** Python | Pandas | MySQL Workbench | SQL | Power Query | Power BI

Part 1: SQL Business Analysis

Analyze retail data and answer ten business questions using MySQL

Analysis Areas

Customer Demographics and Loyalty

Monthly Returns and Sales Growth

Employee and Store Performance

Payment Methods and Promotion Impact

Price Bands and Basket Analysis

SQL Techniques

- Joins, CTEs, views, aggregations, subqueries, and window functions
- **Tool Used**
  MySQL Workbench

Customer Demographics Analysis

Retrieve the number of unique customers by country and city along with their average purchase value

```sql
USE global_fashion_retail;

CREATE OR REPLACE VIEW vw_q01_customer_demographics
AS
SELECT
    c.Country,
    c.City,
    t.Currency,
    COUNT(DISTINCT t.Customer_ID) AS
unique_customers,
    COUNT(DISTINCT t.Invoice_ID) AS total_purchases,
    ROUND(SUM(t.Line_Total), 2) AS total_revenue,
    ROUND(
        SUM(t.Line_Total) /
        NULLIF(COUNT(DISTINCT t.Invoice_ID), 0),
        2
    ) AS average_purchase_value
```

```sql
FROM vw_clean_transactions t
INNER JOIN vw_clean_customers c
    ON t.Customer_ID =
c.Customer_ID
WHERE t.Transaction_Type = 'Sale'
GROUP BY
    c.Country,
    c.City,
    t.Currency;
```

display the results query:

```sql
SELECT *
FROM vw_q01_customer_demographics
ORDER BY total_revenue DESC
LIMIT 100;
```

Result

Country	City	Currency	unique_customers	total_purchases	total_revenue	average_purchase_value
中国	深圳	CNY	55915	197387	106472651.29	539.41
中国	上海	CNY	40968	194139	104904275.27	540.36
中国	广州	CNY	32994	184615	100887044.29	546.47
中国	北京	CNY	46204	152088	83184171.51	546.95
中国	重庆	CNY	23998	102760	56136353.86	546.29
United States	New York	USD	48555	239515	16672572.05	69.61
United States	Los Angeles	USD	43904	228362	15915704.25	69.7



Business Insight

- Shenzhen has the largest customer base among the displayed cities, with 55,915 unique customers.
- New York records 239,515 purchases, the highest among the displayed US cities.
- High-performing cities can be prioritized for inventory planning, targeted marketing, and future expansion.

Monthly Sales vs Returns Trends

Calculate the monthly return percentage compared with total transactions and examine how the return rate changes over time

```sql
CREATE OR REPLACE VIEW vw_q02_monthly_returns AS
SELECT
    CAST(
        DATE_FORMAT(Transaction_Date, '%Y-%m-01')
        AS DATE
    ) AS month_start,

    COUNT(DISTINCT Invoice_ID) AS total_transactions,

    COUNT(
        DISTINCT CASE
            WHEN Transaction_Type = 'Sale'
            THEN Invoice_ID
        END
    ) AS sales_transactions,

    COUNT(
        DISTINCT CASE
            WHEN Transaction_Type = 'Return'
            THEN Invoice_ID
        END
    ) AS return_transactions,
    ROUND(
        100.0 *
        COUNT(
            DISTINCT CASE
                WHEN Transaction_Type =
'Return'
                THEN Invoice_ID
            END
        ) /
        NULLIF(COUNT(DISTINCT Invoice_ID), 0),
        2
    ) AS return_percentage

FROM vw_clean_transactions
GROUP BY
    CAST(
        DATE_FORMAT(Transaction_Date, '%Y-%m-
01')
        AS DATE
    );
```

display the results query:

```sql
SELECT *
FROM vw_q02_monthly_returns
ORDER BY month_start;
```

Result

month_start	total_transactions	sales_transactions	return_transactions	return_percentage
2023-01-01	121713	116153	5560	4.57
2023-02-01	84221	80372	3849	4.57
2023-03-01	186529	177969	8560	4.59
2023-04-01	132356	126360	5996	4.53
2023-05-01	150629	143884	6745	4.48
2023-06-01	93653	89464	4189	4.47



Business Insight

- An increasing monthly return percentage may indicate product-quality, sizing, fulfilment or customer-expectation problems.
- Months with unusually high return rates should be investigated by category, product and store.

Top-Performing Categories

Identify the highest-revenue product categories and calculate their percentage contribution to sales

```sql
CREATE OR REPLACE VIEW
vw_q03_category_performance AS
SELECT
    p.Category,
    t.Currency,

    ROUND(SUM(t.Line_Total), 2) AS
total_revenue,

    ROUND(
        100.0 * SUM(t.Line_Total)
        / SUM(SUM(t.Line_Total)) OVER (
            PARTITION BY t.Currency
        ),
        2
    ) AS revenue_share_percentage,
```

```sql
    DENSE_RANK() OVER (
        PARTITION BY t.Currency
        ORDER BY SUM(t.Line_Total)
DESC
    ) AS category_rank

FROM vw_clean_transactions t

INNER JOIN vw_clean_products p
    ON t.Product_ID = p.Product_ID

WHERE t.Transaction_Type = 'Sale'

GROUP BY
    p.Category,
    t.Currency;
```

Display all category results

```sql
SELECT *
FROM vw_q03_category_performance
ORDER BY Currency, category_rank;
```

Result

Category	Currency	total_revenue	revenue_share_percentage	category_rank
Masculine	CNY	263888180.41	46.27	1
Feminine	CNY	255817211.52	44.86	2
Children	CNY	50602034.24	8.87	3
Feminine	EUR	60291312.11	55.9	1
Masculine	EUR	37931941.34	35.17	2

Business Insight



- The highest-ranked categories are the main revenue drivers.
- The company should maintain sufficient inventory for these categories while reviewing pricing and promotional strategies for categories with a low revenue contribution.

Payment Method Performance

Compare transaction volume, quantity sold, revenue and average basket value across payment methods

```sql
CREATE OR REPLACE VIEW
vw_q04_payment_performance AS
SELECT
    Payment_Method,
    Currency,

    COUNT(DISTINCT Invoice_ID) AS
total_baskets,

    SUM(Quantity) AS total_items_sold,

    ROUND(SUM(Line_Total), 2) AS
total_revenue,

    ROUND(
        SUM(Line_Total) /
        NULLIF(COUNT(DISTINCT Invoice_ID),
0),
        2
    ) AS average_basket_value,
    ROUND(
        SUM(Quantity) /
        NULLIF(COUNT(DISTINCT Invoice_ID), 0),
        2
    ) AS average_items_per_basket

FROM vw_clean_transactions

WHERE Transaction_Type = 'Sale'

GROUP BY
    Payment_Method,
    Currency;
```

Display the results

```sql
SELECT *
FROM vw_q04_payment_performance
ORDER BY Currency, total_revenue DESC;
```

Result

Payment_Method	Currency	total_baskets	total_items_sold	total_revenue	average_basket_value	average_items_per_basket
Credit Card	CNY	841028	1297093	456951277.8	543.32	1.54
Cash	CNY	208795	321867	113356148.37	542.91	1.54
Credit Card	EUR	1378421	2123706	86227579.45	62.56	1.54
Cash	EUR	344822	532078	21627289.29	62.72	1.54
Credit Card	GBP	338355	521353	14714781.35	43.49	1.54
Cash	GBP	85059	131669	3719240.65	43.73	1.55



Business Insight

- A payment method may generate the highest number of transactions while another produces a higher average basket value.
- These findings can help the business improve payment partnerships and customer checkout options.

Promotion Impact Analysis

Compare sales volume and revenue for promotional sales against non-promotional sales

```sql
CREATE OR REPLACE VIEW vw_q05_promotion_impact AS
SELECT
    CASE
        WHEN Discount > 0 THEN 'Promotional'
        ELSE 'Non-Promotional'
    END AS promotion_status,

    Currency,

    COUNT(DISTINCT Invoice_ID) AS total_baskets,

    SUM(Quantity) AS total_items_sold,

    COUNT(DISTINCT DATE(Transaction_Date)) AS
active_sales_days,

    ROUND(
        AVG(Discount) * 100,
        2
    ) AS average_discount_percentage,

    ROUND(
        SUM(Unit_Price * Quantity),
        2
    ) AS gross_sales_value,

    ROUND(
        SUM(Line_Total),
        2
    ) AS net_revenue,

    ROUND(
        SUM(Line_Total) /
        NULLIF(COUNT(DISTINCT Invoice_ID), 0),
        2
    ) AS average_basket_value,

    ROUND(
        SUM(Line_Total) /
        NULLIF(COUNT(DISTINCT DATE(Transaction_Date)), 0),
        2
    ) AS average_daily_revenue

FROM vw_clean_transactions

WHERE Transaction_Type = 'Sale'

GROUP BY
    CASE
        WHEN Discount > 0 THEN 'Promotional'
        ELSE 'Non-Promotional'
    END,
    Currency;
```

Display the results

```sql
SELECT *
FROM vw_q05_promotion_impact
ORDER BY Currency, promotion_status;
```

Result

promotion_status	Currency	total_baskets	total_items_sold	active_sales_days	average_discount_percentage	gross_sales_value	net_revenue	average_basket_value	average_daily_revenue
Non-Promotional	CNY	767120	1155708	782	0	465290659	465290659	606.54	595000.84
Promotional	CNY	318200	463252	174	42.2	184516899.5	105016767.17	330.03	603544.64
Non-Promotional	EUR	1235116	1855996	782	0	86962634.7	86962634.7	70.41	111205.42
Promotional	EUR	546780	799788	174	42.75	37002018.3	20892234.05	38.21	120070.31
Non-Promotional	GBP	300615	452811	782	0	14837986	14837986	49.36	18974.41
Promotional	GBP	136555	200211	174	43.33	6415290	3596036	26.33	20666.87
Non-Promotional	USD	833424	1253477	782	0	64813412.5	64813412.5	77.77	82881.6

Business Insight



- Promotions are effective when they increase sales volume and revenue without causing an excessive decline in average basket value or profit margin.
- Weak promotions should be redesigned or targeted toward selected categories.

If the discount table contains category-specific promotions, include category or subcategory in the join to avoid matching an unrelated promotion.

Customer Loyalty Tracking

Measure the proportion of new and repeat customers each month and compare their average spending

```sql
CREATE OR REPLACE VIEW vw_q06_customer_loyalty AS

WITH invoice_sales AS
(
    SELECT
        Invoice_ID,
        Customer_ID,
        Currency,

        CAST(
            DATE_FORMAT(Transaction_Date, '%Y-%m-01')
            AS DATE
        ) AS month_start,

        SUM(Line_Total) AS invoice_revenue

    FROM vw_clean_transactions

    WHERE Transaction_Type = 'Sale'

    GROUP BY
        Invoice_ID,
        Customer_ID,
        Currency,
        CAST(
            DATE_FORMAT(Transaction_Date, '%Y-%m-01')
            AS DATE
        )
),
```

```sql
customer_monthly AS
(
    SELECT
        month_start,
        Customer_ID,
        Currency,
        SUM(invoice_revenue) AS
monthly_customer_spend

    FROM invoice_sales

    GROUP BY
        month_start,
        Customer_ID,
        Currency
),

first_purchase AS
(
    SELECT
        Customer_ID,
        MIN(month_start) AS
first_purchase_month

    FROM customer_monthly

    GROUP BY Customer_ID
),
```

```sql
classified_customers AS
(
    SELECT
        cm.month_start,
        cm.Customer_ID,
        cm.Currency,
        cm.monthly_customer_spend,

        CASE
            WHEN cm.month_start =
fp.first_purchase_month
                THEN 'New'
            ELSE 'Repeat'
        END AS customer_type

    FROM customer_monthly cm

    INNER JOIN first_purchase fp
        ON cm.Customer_ID = fp.Customer_ID
)

SELECT
    month_start,
    Currency,
    customer_type,

    COUNT(*) AS customer_count,

    ROUND(
        100.0 * COUNT(*) /
        SUM(COUNT(*)) OVER (
            PARTITION BY month_start, Currency
        ),
        2
    ) AS customer_percentage,
```

```
    ROUND(
        AVG(monthly_customer_spend),
        2
    ) AS average_customer_spend,

    ROUND(
        SUM(monthly_customer_spend),
        2
    ) AS total_customer_revenue

FROM classified_customers

GROUP BY
    month_start,
    Currency,
    customer_type;
```

Display the results:

```
SELECT *
FROM vw_q06_customer_loyalty
ORDER BY month_start, Currency, customer_type;
```

Result

month_start	Currency	customer_type	customer_count	customer_percentage	average_customer_spend	total_customer_revenue
2023-01-01	CNY	New	24045	100.00	431.2	10368137.9
2023-01-01	EUR	New	44509	100.00	51.24	2280575.15
2023-01-01	GBP	New	9626	100.00	38.43	369971
2023-01-01	USD	New	31234	100.00	56.82	1774855.6
2023-02-01	CNY	New	16210	88.37	476.24	7719818.5
2023-02-01	CNY	Repeat	2134	11.63	490.51	1046739



Business Insight

A growing repeat-customer base indicates stronger customer retention. If repeat customers spend more than new customers, loyalty programs and personalized offers could provide additional value.

Store Performance Ranking

Identify the top and bottom stores based on sales revenue and return rate

```sql
CREATE OR REPLACE VIEW vw_q07_store_performance AS
SELECT
    s.Store_ID,
    s.Store_Name,
    s.Country,
    s.City,
    t.Currency,

    ROUND(
        SUM(
            CASE
                WHEN t.Transaction_Type = 'Sale'
                THEN t.Line_Total
                ELSE 0
            END
        ),
        2
    ) AS total_revenue,

    ROUND(
        100.0 *
        COUNT(
            DISTINCT CASE
                WHEN t.Transaction_Type = 'Return'
                THEN t.Invoice_ID
            END
        ) /
        NULLIF(COUNT(DISTINCT t.Invoice_ID), 0),
        2
    ) AS return_rate

FROM vw_clean_transactions t

JOIN vw_clean_stores s
    ON t.Store_ID = s.Store_ID

GROUP BY
    s.Store_ID,
    s.Store_Name,
    s.Country,
    s.City,
    t.Currency;
```

Display all store results:

```sql
SELECT *
FROM vw_q07_store_performance
WHERE Currency = 'USD'
ORDER BY total_revenue DESC
LIMIT 5;
```

```sql
SELECT *
FROM vw_q07_store_performance
WHERE Currency = 'USD'
ORDER BY total_revenue ASC
LIMIT 5;
```

```sql
SELECT *
FROM vw_q07_store_performance
WHERE Currency = 'USD'
ORDER BY return_rate DESC
LIMIT 5;
```

Result

Store_ID	Store_Name	Country	City	Currency	total_revenue	return_rate
5	Store Phoenix	United States	Phoenix	USD	9591131.42	4.47
3	Store Chicago	United States	Chicago	USD	11238559.05	4.51
4	Store Houston	United States	Houston	USD	14730576.15	4.39
2	Store Los Angeles	United States	Los Angeles	USD	21424373.36	4.53
1	Store New York	United States	New York	USD	22479565.75	4.47



Business Insight

- High-revenue stores with low return rates can be used as performance benchmarks.
- Stores with weak revenue or high return rates may require changes in product assortment, employee training or operational processes.

Price-Band Contribution

Divide products into price bands and measure their contribution to revenue and returns

```sql
CREATE OR REPLACE VIEW vw_q08_price_bands AS

WITH price_summary AS
(
    SELECT
        Currency,

        CASE
            WHEN Unit_Price < 25 THEN 'Low: Below 25'
            WHEN Unit_Price < 50 THEN 'Medium: 25-49'
            WHEN Unit_Price < 100 THEN 'High: 50-99'
            ELSE 'Premium: 100+'
        END AS price_band,

        SUM(
            CASE
                WHEN Transaction_Type = 'Sale'
                THEN Line_Total
                ELSE 0
            END
        ) AS sales_revenue,

        ABS(
            SUM(
                CASE
                    WHEN Transaction_Type = 'Return'
                    THEN Line_Total
                    ELSE 0
                END
            )
        ) AS returned_value

    FROM vw_clean_transactions

    GROUP BY
        Currency,
        price_band
)
```

```sql
SELECT
    Currency,
    price_band,
    ROUND(sales_revenue, 2) AS sales_revenue,
    ROUND(returned_value, 2) AS returned_value,

    ROUND(
        100 * sales_revenue /
        SUM(sales_revenue) OVER (PARTITION BY Currency),
        2
    ) AS revenue_share_percentage,

    ROUND(
        100 * returned_value /
        NULLIF(
            SUM(returned_value) OVER (PARTITION BY Currency),
            0
        ),
        2
    ) AS return_share_percentage

FROM price_summary;
```

Display the results

```sql
SELECT *
FROM vw_q08_price_bands
ORDER BY Currency, sales_revenue DESC;
```

Result

Currency	price_band	sales_revenue	returned_value	revenue_share_percentage	return_share_percentage
CNY	Medium: 25-49	264804.13	14042.96	0.05	0.04
EUR	High: 50-99	50519348.16	2848857.29	46.84	46.92
EUR	Medium: 25-49	39736893.36	2240866.75	36.84	36.9
EUR	Premium: 100+	10251010.12	570446.06	9.5	9.39
EUR	Low: Below 25	7347617.1	411946.21	6.81	6.78
GBP	Medium: 25-49	8756598.59	489654.74	47.5	47.32



Business Insight

- The result explains whether customers mainly purchase low-, medium-, high- or premium-priced products.
- A price band with high revenue but also high returned value may require product-quality or pricing investigation.

Basket-Size Insights

Calculate the average number of items and average revenue per sales basket for each country

```sql
CREATE OR REPLACE VIEW vw_q09_basket_country
AS

WITH baskets AS
(
    SELECT
        s.Country,
        t.Currency,
        t.Invoice_ID,
        SUM(t.Quantity) AS basket_items,
        SUM(t.Line_Total) AS basket_revenue

    FROM vw_clean_transactions t

    JOIN vw_clean_stores s
        ON t.Store_ID = s.Store_ID

    WHERE t.Transaction_Type = 'Sale'

    GROUP BY
        s.Country,
        t.Currency,
        t.Invoice_ID
)
```

```sql
SELECT
    Country,
    Currency,
    COUNT(*) AS total_baskets,

    ROUND(
        AVG(basket_items),
        2
    ) AS average_items_per_basket,

    ROUND(
        AVG(basket_revenue),
        2
    ) AS average_revenue_per_basket,

    ROUND(
        SUM(basket_revenue),
        2
    ) AS total_revenue

FROM baskets

GROUP BY
    Country,
    Currency;
```

Display the results

```sql
SELECT *
FROM vw_q09_basket_country
ORDER BY Currency, average_revenue_per_basket
DESC;
```

Result

Country	Currency	total_baskets	average_items_per_basket	average_revenue_per_basket	total_revenue
中国	CNY	1049823	1.54	543.24	570307426.17
España	EUR	386278	1.54	62.78	24251259.52
Portugal	EUR	392354	1.54	62.72	24609479.69
France	EUR	438134	1.54	62.49	27378649.27
Deutschland	EUR	506477	1.54	62.42	31615480.26

Business Insight



- Countries with larger basket sizes offer opportunities for cross-selling and product bundles.
- In the observed result, China recorded an average basket revenue of approximately CNY 543.24, while the United States recorded approximately USD 69.66.
- Values in different currencies should not be compared directly without currency conversion.

Monthly Revenue Growth

Calculate month-over-month revenue growth overall and for the top three product categories

```sql
CREATE OR REPLACE VIEW vw_q10_monthly_growth AS

WITH monthly AS
(
    SELECT
        DATE_FORMAT(
            t.Transaction_Date,
            '%Y-%m-01'
        ) AS month_start,

        t.Currency,
        p.Category,
        SUM(t.Line_Total) AS revenue

    FROM vw_clean_transactions t

    JOIN vw_clean_products p
        ON t.Product_ID = p.Product_ID

    WHERE t.Transaction_Type = 'Sale'

    GROUP BY
        month_start,
        t.Currency,
        p.Category
),
```

```sql
combined AS
(
    SELECT
        month_start,
        Currency,
        'Overall' AS revenue_group,
        SUM(revenue) AS revenue

    FROM monthly

    GROUP BY
        month_start,
        Currency

    UNION ALL

    SELECT
        m.month_start,
        m.Currency,
        m.Category AS revenue_group,
        m.revenue

    FROM monthly m

    JOIN vw_q03_category_performance q
        ON m.Category = q.Category
       AND m.Currency = q.Currency

    WHERE q.category_rank <= 3
),
```

```sql
growth AS
(
    SELECT
        month_start,
        Currency,
        revenue_group,
        revenue,

        LAG(revenue) OVER (
            PARTITION BY Currency,
revenue_group
            ORDER BY month_start
        ) AS previous_revenue

    FROM combined
)

SELECT
    month_start,
    Currency,
    revenue_group,

    ROUND(revenue, 2) AS revenue,

    ROUND(
        100.0 * (revenue -
previous_revenue) /
        NULLIF(previous_revenue, 0),
        2
    ) AS monthly_growth_percentage

FROM growth;
```

Check the results

```
SELECT *
FROM vw_q10_monthly_growth
ORDER BY Currency, revenue_group,
month_start;
```

Result

month_start	Currency	revenue_group	revenue	monthly_growth_percentage
2023-01-01	CNY	Children	1030476.5	NULL
2023-02-01	CNY	Children	636482.5	-38.23
2023-03-01	CNY	Children	2746382.19	331.49
2023-04-01	CNY	Children	1242932	-54.74
2023-05-01	CNY	Children	1861443.14	49.76
2023-06-01	CNY	Children	1064362	-42.82

Business Insight



- Positive growth means revenue increased compared with the previous month, while negative growth identifies periods requiring investigation.
- Category-level growth helps management determine which products are driving or reducing overall performance.

Part 2: Interactive Dashboards in Power BI

Convert the cleaned retail data into interactive dashboards for performance monitoring and detailed business analysis

Dashboard Areas

- Sales and Returns Overview
- Geographic Insights
- Category and Product Insights
- Promotion Effectiveness
- Customer Behaviour

Power BI Features

- DAX measures, KPI cards, slicers, drill-downs, interactive charts, and data modelling
- **Tools Used**
  Power BI Desktop, Power Query, DAX, and MySQL ODBC Connector

Data Cleaning and Transformation

Connected MySQL to Power BI using ODBC.

Cleaned approximately **6.42 million transaction records**.

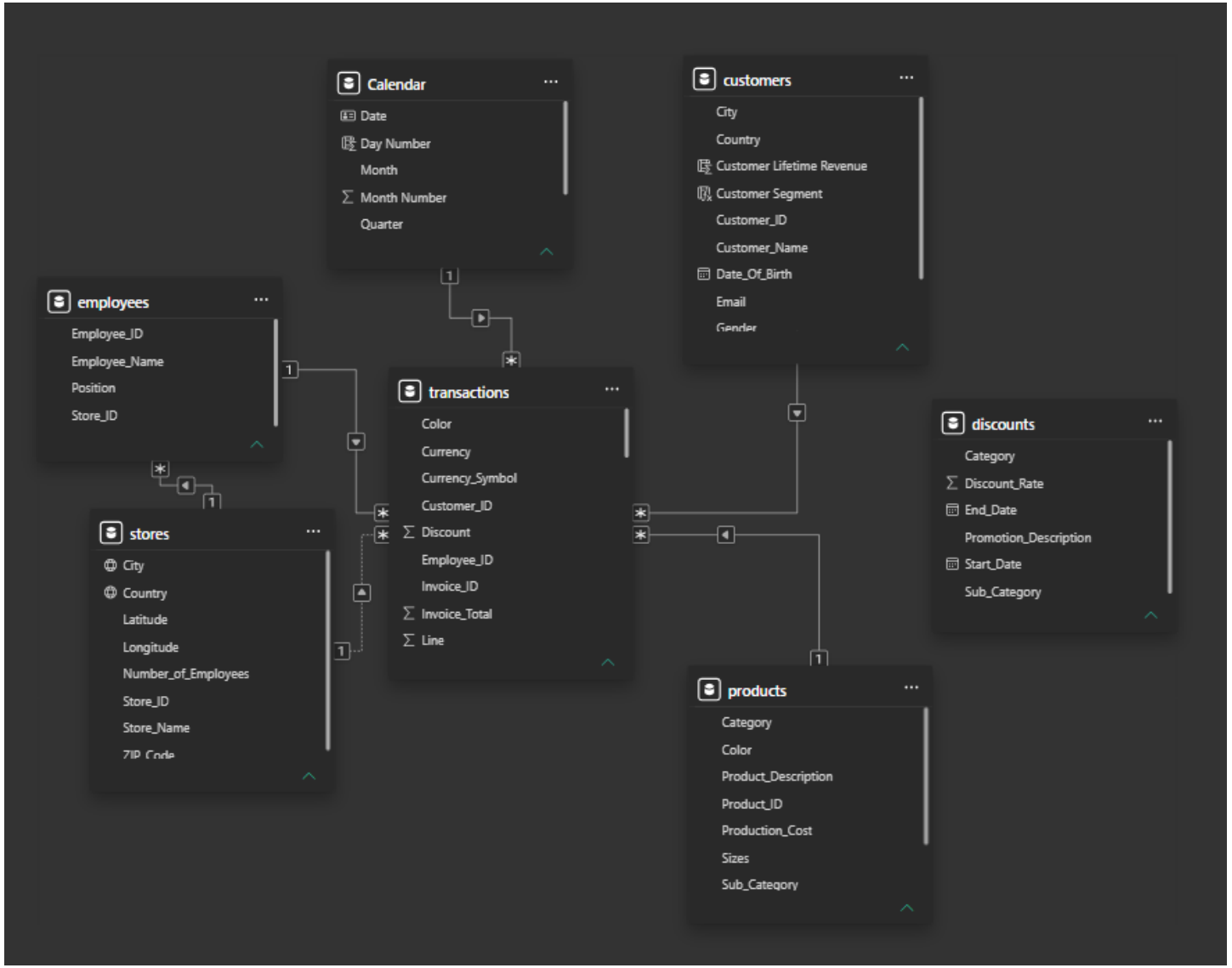
Removed duplicates and handled missing values.

Corrected data types and standardized categories.

Created Calendar table and DAX measures.

- **Outcome:** Reliable, analysis-ready data for interactive dashboards.

Data Model View



Data Model and Relationships

Tables used:

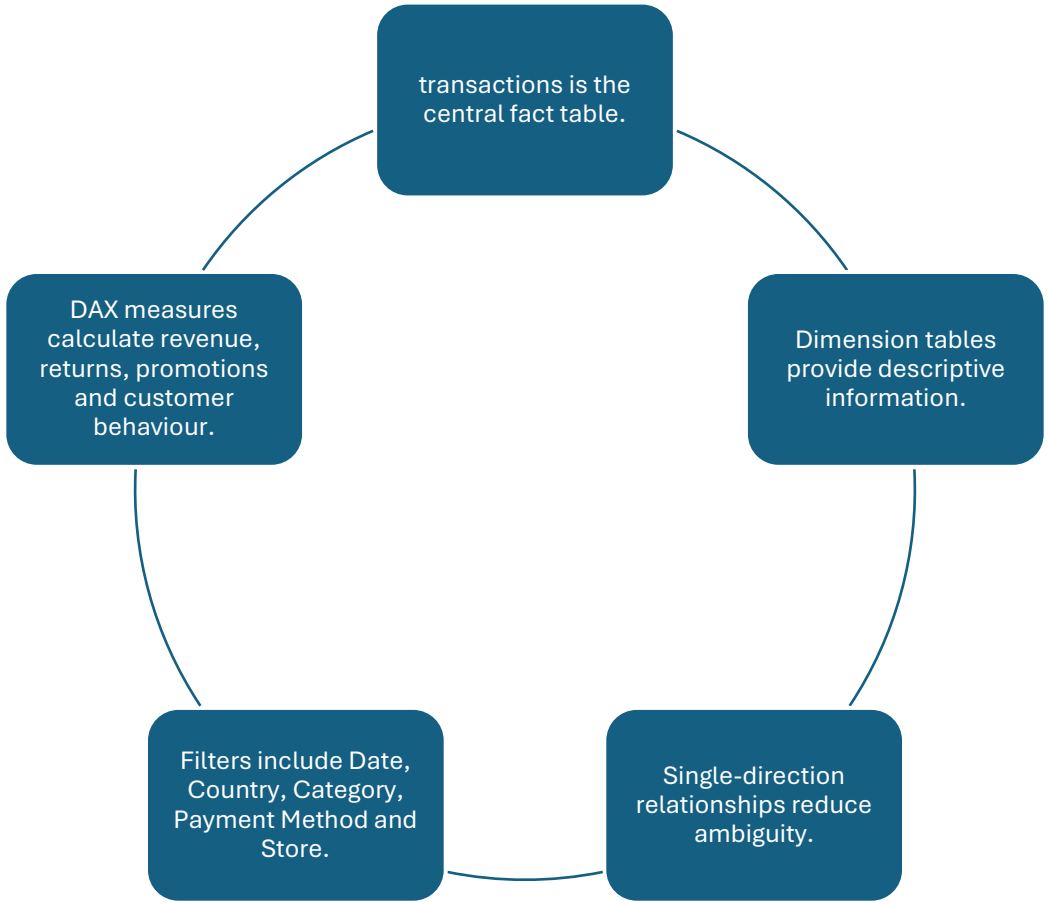
- Transactions
- Customers
- Products
- Stores
- Employees
- Discounts and Calendar

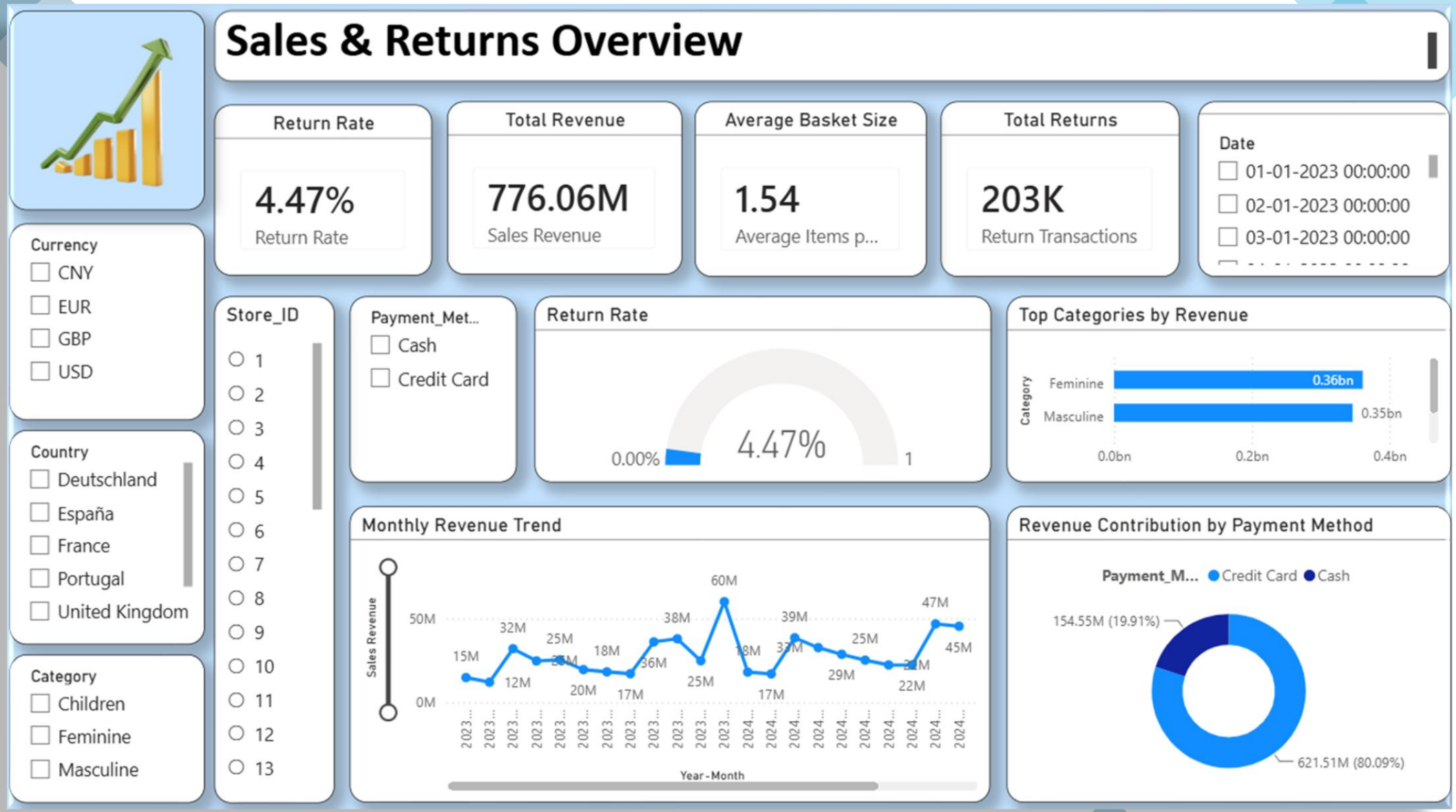
Relationships:

- Customers 1:* Transactions
- Products 1:* Transactions
- Stores 1:* Transactions
- Employees 1:* Transactions
- Calendar 1:* Transactions

The model uses a star-schema structure.

Model Summary




Sales & Returns Overview
Return Rate
4.47%
Return Rate
Total Revenue
776.06M
Sales Revenue
Average Basket Size
1.54
Average Items p...
Total Returns
203K
Return Transactions
Date
01-01-2023 00:00:00
02-01-2023 00:00:00
03-01-2023 00:00:00
Currency
CNY
EUR
GBP
USD
Country
Deutschland
España
France
Portugal
United Kingdom
Category
Children
Feminine
Masculine
Store_ID
1
2
3
4
5
6
7
8
9
10
11
12
13
Payment_Met...
Cash
Credit Card
Return Rate
0.00%
4.47%
1
Top Categories by Revenue
Category
Feminine
0.36bn
Masculine
0.35bn
0.0bn
0.2bn
0.4bn
Monthly Revenue Trend
Sales Revenue
50M
0M
15M
12M
32M
25M
20M
18M
17M
36M
38M
25M
60M
17M
39M
25M
29M
22M
47M
45M
Year-Month
Revenue Contribution by Payment Method
Payment_M...
Credit Card
Cash
154.55M (19.91%)
621.51M (80.09%)

Sales and Returns Insights

Key insights

- The overall return rate is **4.47%**.
- The dataset contains approximately **53,000 return transactions**.
- Customers purchase an average of **1.54 items per basket**.
- The dashboard covers transactions from **January 2023 to March 2025**.
- Revenue changes noticeably by month, with several peaks visible during the analysis period.
- Category and payment-method visuals identify the largest contributors to sales.

Executive interpretation

A 4.47% return rate requires continued monitoring at category and product levels.
The low average basket size suggests an opportunity for product bundles and cross-selling.

Geographic Insights

Payment_Method
- Cash
- Credit Card

Unique Customers
1M
Unique Customers

Average Spend per Customer
604.55
Average Spend per Custo...

Date
- 01-01-2023 00:00:00
- 02-01-2023 00:00:00
- 03-01-2023 00:00:00

Currency
- CNY
- EUR
- GBP
- USD

Category
- Children
- Feminine
- Masculine

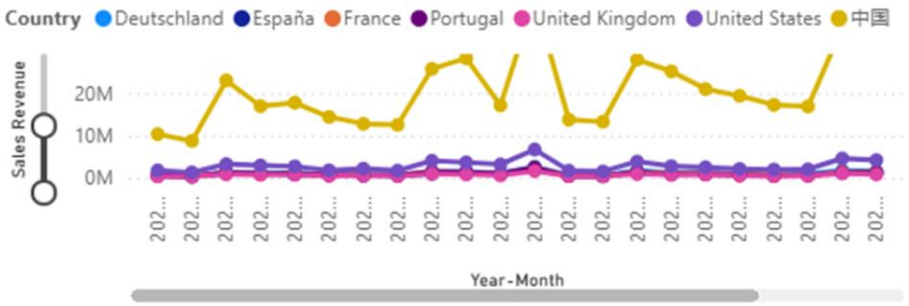
Country
- Deutschl...
- España
- France
- Portugal
- United K...

Store_ID
- 1
- 2
- 3
- 4
- 5
- 6

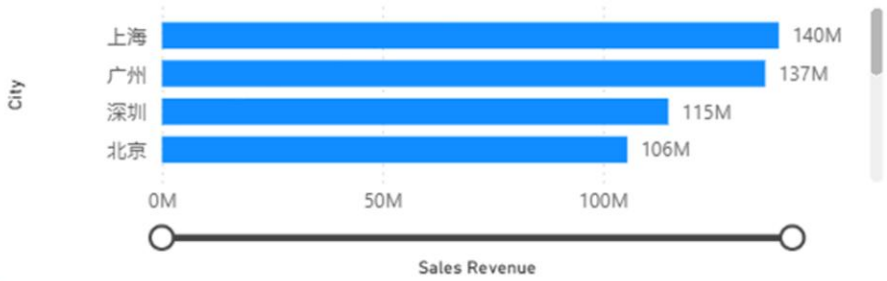
Revenue by Country



Revenue Trend by Country



Top Cities by Revenue



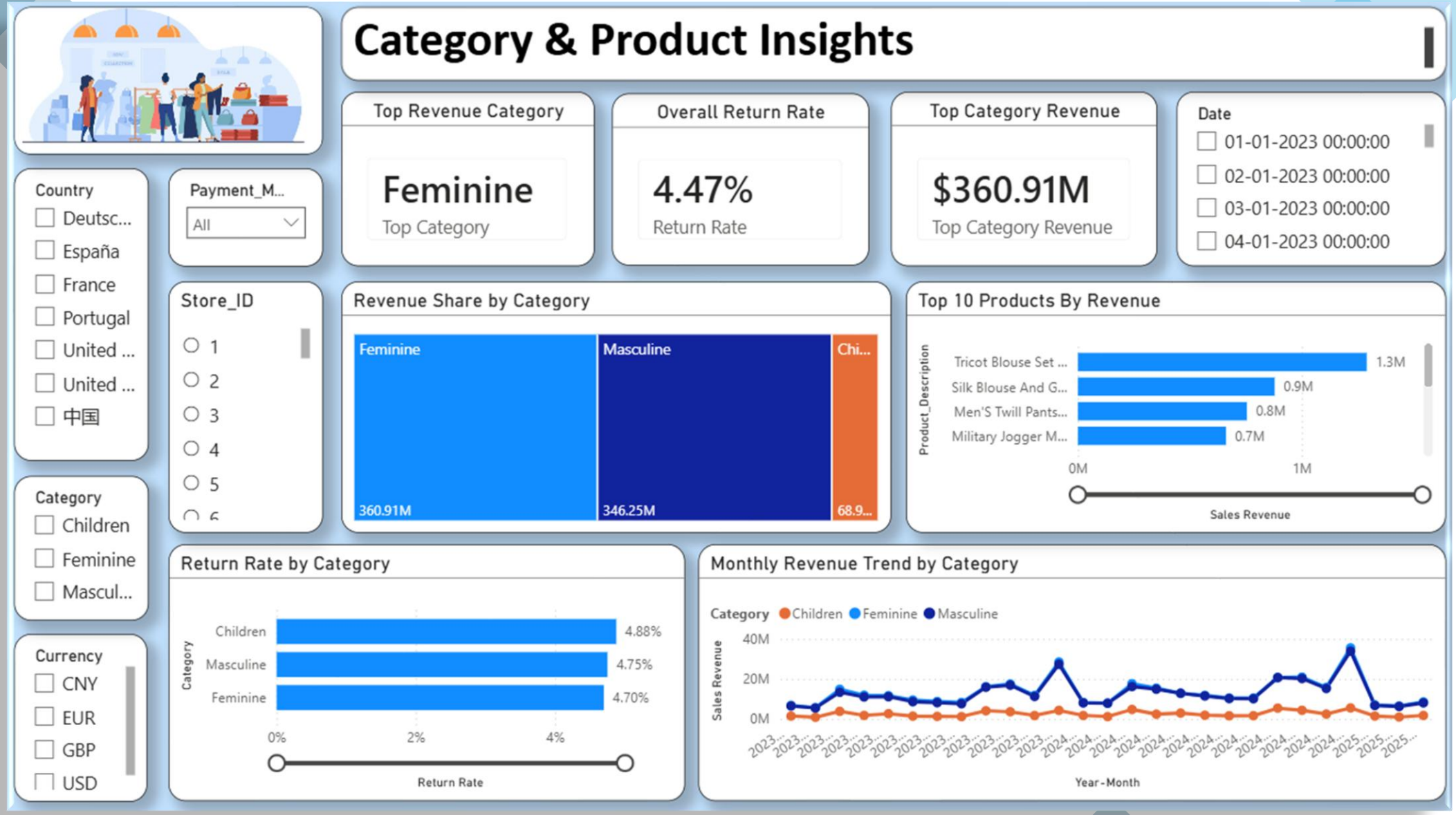
Country Performance Summary

Country	Sales Revenue	Average Spend per Customer	Unique Customers
中国	57,03,07,426.17	1958.82	291149
United States	7,94,64,205.74	262.37	302867
Deutschland	3,16,15,480.26	194.83	162272
France	2,73,78,649.27	181.80	150595
Portugal	2,46,09,479.69	241.39	101950
España	2,42,51,259.52	164.40	147513
Total	**77,60,60,522.65**	**604.55**	**1283707**

Geographic Insights

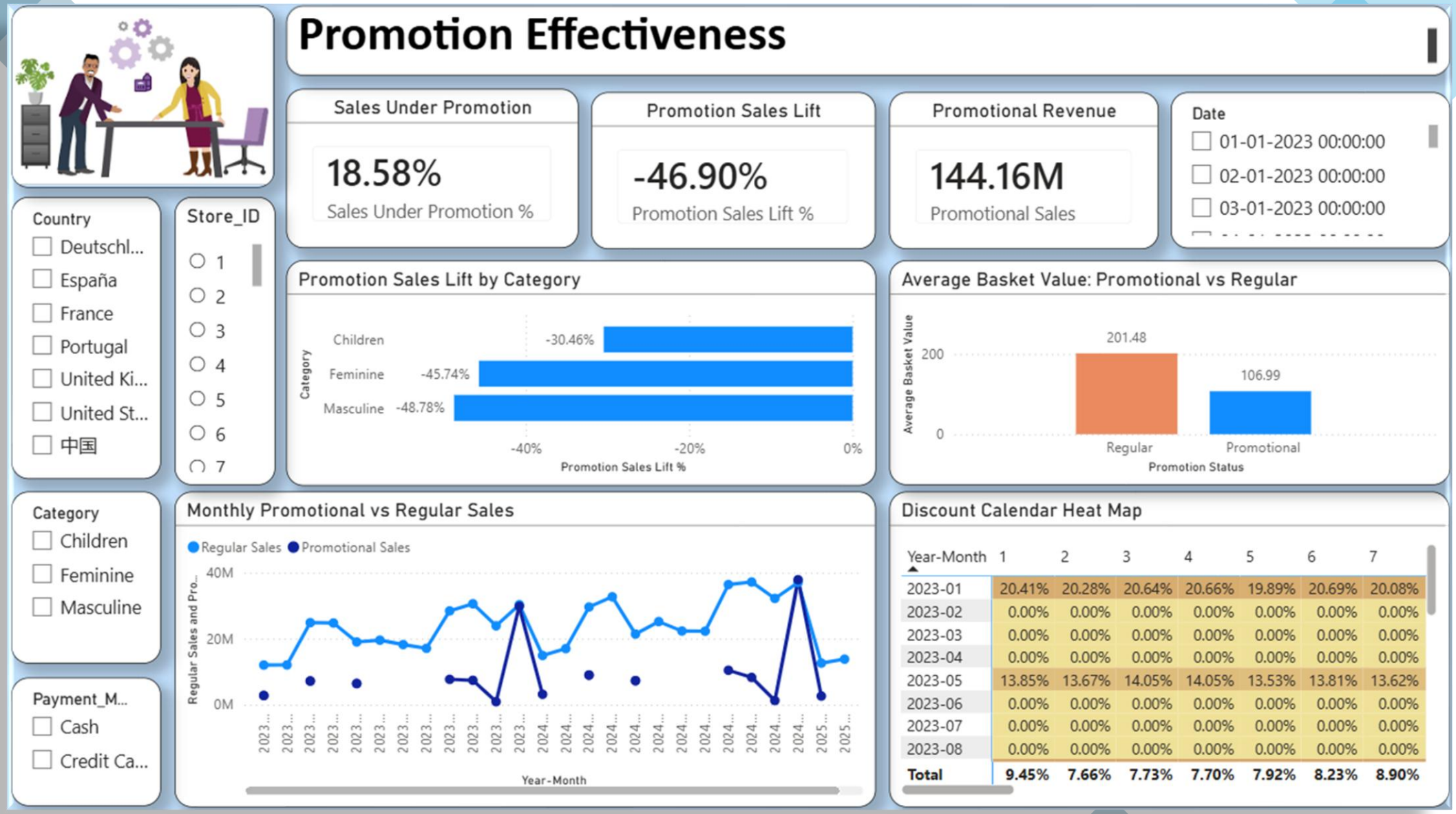
- The business served approximately **1.28M unique customers**.
- Average spending per customer was **604.55**.
- China dominated country revenue and accounted for most of the highest-revenue cities.
- The United States generated approximately **79.95M**, with **302,687 customers** and average spending of **262.37**.
- China’s monthly revenue remained significantly higher than the other countries.




Category & Product Insights
Top Revenue Category
Feminine
Top Category
Overall Return Rate
4.47%
Return Rate
Top Category Revenue
$360.91M
Top Category Revenue
Date
01-01-2023 00:00:00
02-01-2023 00:00:00
03-01-2023 00:00:00
04-01-2023 00:00:00
Country
Deutsc...
España
France
Portugal
United ...
United ...
中国
Payment_M...
All
Store_ID
1
2
3
4
5
Revenue Share by Category
Feminine
360.91M
Masculine
346.25M
Chi...
68.9...
Top 10 Products By Revenue
Product_Description
Tricot Blouse Set ...
1.3M
Silk Blouse And G...
0.9M
Men'S Twill Pants...
0.8M
Military Jogger M...
0.7M
0M
1M
Sales Revenue
Category
Children
Feminine
Mascul...
Currency
CNY
EUR
GBP
USD
Return Rate by Category
Category
Children
4.88%
Masculine
4.75%
Feminine
4.70%
0%
2%
4%
Return Rate
Monthly Revenue Trend by Category
Category
Children
Feminine
Masculine
Sales Revenue
40M
20M
0M
Year-Month

Category & Product Insights

- Feminine was the leading category with **360.91M** in revenue.
- Masculine generated **346.25M**, while Children generated approximately **68.9M**.
- Feminine and Masculine together contributed approximately **91.1%** of category revenue.
- Children recorded the highest return rate at **4.88%**, followed by Masculine at **4.75%** and Feminine at **4.70%**.
- Revenue trends show similar performance between Feminine and Masculine across most months.


Promotion Effectiveness
Sales Under Promotion
18.58%
Sales Under Promotion %
Promotion Sales Lift
-46.90%
Promotion Sales Lift %
Promotional Revenue
144.16M
Promotional Sales
Date
01-01-2023 00:00:00
02-01-2023 00:00:00
03-01-2023 00:00:00
Country
Deutschl...
España
France
Portugal
United Ki...
United St...
中国
Store_ID
1
2
3
4
5
6
7
Promotion Sales Lift by Category
Children
Feminine
Masculine
-30.46%
-45.74%
-48.78%
Category
-40%
-20%
0%
Promotion Sales Lift %
Average Basket Value: Promotional vs Regular
Average Basket Value
201.48
106.99
200
0
Regular
Promotional
Promotion Status
Category
Children
Feminine
Masculine
Monthly Promotional vs Regular Sales
Regular Sales
Promotional Sales
40M
20M
0M
Regular Sales and Pro...
Year-Month
Discount Calendar Heat Map
Year-Month 1 2 3 4 5 6 7
2023-01 20.41% 20.28% 20.64% 20.66% 19.89% 20.69% 20.08%
2023-02 0.00% 0.00% 0.00% 0.00% 0.00% 0.00% 0.00%
2023-03 0.00% 0.00% 0.00% 0.00% 0.00% 0.00% 0.00%
2023-04 0.00% 0.00% 0.00% 0.00% 0.00% 0.00% 0.00%
2023-05 13.85% 13.67% 14.05% 14.05% 13.53% 13.81% 13.62%
2023-06 0.00% 0.00% 0.00% 0.00% 0.00% 0.00% 0.00%
2023-07 0.00% 0.00% 0.00% 0.00% 0.00% 0.00% 0.00%
2023-08 0.00% 0.00% 0.00% 0.00% 0.00% 0.00% 0.00%
Total 9.45% 7.66% 7.73% 7.70% 7.92% 8.23% 8.90%
Payment_M...
Cash
Credit Ca...

Promotion Effectiveness Insights

- Promotional transactions contributed **18.58%** of total sales.
- Promotional revenue reached **144.16M**.
- Promotional average basket value was **106.99**, compared with **201.48** for regular sales.
- Promotion sales lift was **–46.90%**, indicating that discounted baskets generated lower revenue.
- Promotion lift was negative across all categories: Masculine **–48.78%**, Feminine **–45.74%**, and Children **–30.46%**.
- Children responded better to promotions than the other categories, although its lift remained negative.




Customer Behavior
New Customer Rate
24.92%
New Customer %
Average Order Value
178.93
Average Basket Va...
Repeat Customer Rate
75.08%
Repeat Customer %
Date
01-01-2023 00:00:00
02-01-2023 00:00:00
03-01-2023 00:00:00
Country
Deutsch...
España
France
Portugal
United K...
United S...
Store_ID
1
2
3
4
5
6
Average Basket Value by Customer Segment
Average Basket Value
500
0
High Value
Medium Value
Low Value
Customer Segment
Average Basket Size by Customer Segment
Customer Segment
Medium Value
High Value
Low Value
1.73
1.65
1.46
0.0
0.5
1.0
1.5
2.0
Average Items per Basket
Category
Children
Feminine
Masculine
Payment_...
All
Customer Segment Distribution
Customer Seg...
Low Value
High Value
Medium Value
126K (9.85%)
181K (14.08%)
976K (76.07%)
Monthly New vs Repeat Customer Trend
Repeat Customer %
New Customer %
100%
50%
0%
Repeat Custome...
Year-Month

Customer Behavior Insights



Repeat customers represented **75.08%**, while new customers represented **24.92%**.



Average order value was **178.93**.



Low-value customers formed the largest segment with **976K customers (76.07%)**.



High-value customers accounted for **181K (14.08%)**, while medium-value customers accounted for **126K (9.85%)**.



Medium-value customers had the largest basket size at **1.73 items**, followed by high-value customers at **1.65** and low-value customers at **1.46**.



New and repeat customer proportions remained relatively stable across the period.

Summary Insights Report

- The business generated **776.06M in revenue**, with an overall **return rate of 4.47%** and an average basket size of **1.54 items**.
- **Credit cards contributed 80.09% of revenue**, making them the dominant payment method.
- The customer base reached approximately **1.28M**, with an average spend of **604.55 per customer**. China was the strongest geographic market.
- The **Feminine category led with 360.91M in revenue**, followed by Masculine at **346.25M**. Together, they contributed approximately **91% of total revenue**.



- Promotional sales generated **144.16M**, but the recorded promotion lift was **–46.90%**, indicating that the discount strategy needs improvement.
- **Repeat customers represented 75.08%** of customers, while new customers accounted for **24.92%**, showing strong customer retention.

- **Final Recommendation**
- Focus on the highest-performing countries and categories, redesign promotions to improve basket value, reduce category-level returns, and use loyalty programs to retain existing customers while attracting more new customers.




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